

# VESSEL ANALYSIS REPORT

| vaf(%) | branch points count | tip point count | vein count | total vein length | average vein length |
|--------|---------------------|-----------------|------------|-------------------|---------------------|
| 1.164  | 50.0                | 165.0           | 155.0      | 3356.275          | 21.653              |

| id | p1.x, p1.y | p2.x, p2.y | length | p1_type | p2_type |
|----|------------|------------|--------|---------|---------|
| 1  | [195,3]    | [178,4]    | 17.243 | TIP     | BRANCH  |
| 2  | [178,4]    | [171,32]   | 29.899 | BRANCH  | BRANCH  |
| 3  | [170,0]    | [178,4]    | 8.243  | TIP     | BRANCH  |
| 4  | [249,1]    | [219,13]   | 34.556 | TIP     | BRANCH  |
| 5  | [194,19]   | [219,13]   | 26.485 | TIP     | BRANCH  |
| 6  | [219,13]   | [276,76]   | 89.711 | BRANCH  | BRANCH  |
| 7  | [171,32]   | [167,34]   | 3.828  | BRANCH  | BRANCH  |
| 8  | [209,56]   | [171,32]   | 50.627 | TIP     | BRANCH  |
| 9  | [167,34]   | [157,45]   | 16.485 | BRANCH  | BRANCH  |
| 10 | [160,20]   | [167,34]   | 20.314 | TIP     | BRANCH  |
| 11 | [156,47]   | [157,45]   | 1.414  | TIP     | BRANCH  |
| 12 | [157,45]   | [120,56]   | 40.971 | BRANCH  | BRANCH  |
| 13 | [346,46]   | [346,53]   | 6.0    | TIP     | BRANCH  |
| 14 | [350,52]   | [346,53]   | 3.414  | TIP     | BRANCH  |
| 15 | [345,64]   | [346,53]   | 12.071 | TIP     | BRANCH  |
| 16 | [9,71]     | [120,56]   | 117.87 | TIP     | BRANCH  |
| 17 | [120,56]   | [135,69]   | 20.142 | BRANCH  | BRANCH  |
| 18 | [135,69]   | [150,71]   | 14.828 | BRANCH  | BRANCH  |
| 19 | [135,69]   | [139,85]   | 17.243 | BRANCH  | BRANCH  |
| 20 | [155,68]   | [150,71]   | 4.828  | TIP     | BRANCH  |
| 21 | [150,71]   | [139,85]   | 20.314 | BRANCH  | BRANCH  |
| 22 | [276,76]   | [295,77]   | 19.243 | BRANCH  | BRANCH  |
| 23 | [280,105]  | [276,76]   | 30.485 | TIP     | BRANCH  |
| 24 | [300,77]   | [295,77]   | 4.0    | TIP     | BRANCH  |
| 25 | [299,86]   | [295,77]   | 9.657  | TIP     | BRANCH  |
| 26 | [139,85]   | [138,94]   | 8.414  | BRANCH  | BRANCH  |

| id | p1.x, p1.y | p2.x, p2.y | length | p1_type | p2_type |
|----|------------|------------|--------|---------|---------|
| 27 | [206,88]   | [201,87]   | 4.414  | TIP     | BRANCH  |
| 28 | [194,105]  | [201,87]   | 19.899 | TIP     | BRANCH  |
| 29 | [200,85]   | [201,87]   | 1.0    | TIP     | BRANCH  |
| 30 | [104,78]   | [100,89]   | 11.657 | TIP     | BRANCH  |
| 31 | [100,89]   | [79,103]   | 25.385 | BRANCH  | BRANCH  |
| 32 | [100,89]   | [105,91]   | 4.414  | BRANCH  | BRANCH  |
| 33 | [132,96]   | [105,91]   | 28.899 | TIP     | BRANCH  |
| 34 | [110,124]  | [105,91]   | 34.071 | TIP     | BRANCH  |
| 35 | [142,95]   | [138,94]   | 3.414  | TIP     | BRANCH  |
| 36 | [136,94]   | [138,94]   | 1.0    | TIP     | BRANCH  |
| 37 | [79,103]   | [65,106]   | 14.243 | BRANCH  | BRANCH  |
| 38 | [79,103]   | [74,120]   | 18.657 | BRANCH  | BRANCH  |
| 39 | [65,106]   | [50,107]   | 14.414 | BRANCH  | BRANCH  |
| 40 | [65,106]   | [74,120]   | 16.728 | BRANCH  | BRANCH  |
| 41 | [50,107]   | [14,171]   | 79.669 | BRANCH  | BRANCH  |
| 42 | [11,94]    | [50,107]   | 43.799 | TIP     | BRANCH  |
| 43 | [74,120]   | [72,137]   | 16.828 | BRANCH  | BRANCH  |
| 44 | [131,117]  | [149,123]  | 19.485 | TIP     | BRANCH  |
| 45 | [150,125]  | [149,123]  | 1.414  | TIP     | BRANCH  |
| 46 | [72,137]   | [83,149]   | 16.728 | BRANCH  | BRANCH  |
| 47 | [72,137]   | [43,197]   | 80.326 | BRANCH  | BRANCH  |
| 48 | [97,130]   | [98,140]   | 9.414  | TIP     | BRANCH  |
| 49 | [143,170]  | [98,140]   | 56.426 | TIP     | BRANCH  |
| 50 | [98,140]   | [83,149]   | 17.314 | BRANCH  | BRANCH  |
| 51 | [87,173]   | [83,149]   | 24.657 | TIP     | BRANCH  |
| 52 | [20,174]   | [14,171]   | 6.243  | TIP     | BRANCH  |
| 53 | [11,180]   | [14,171]   | 8.828  | TIP     | BRANCH  |
| 54 | [54,194]   | [54,196]   | 1.0    | TIP     | BRANCH  |
| 55 | [54,196]   | [53,224]   | 42.042 | BRANCH  | BRANCH  |
| 56 | [54,196]   | [43,197]   | 10.0   | BRANCH  | BRANCH  |
| 57 | [43,197]   | [44,211]   | 14.243 | BRANCH  | BRANCH  |
| 58 | [44,211]   | [32,212]   | 11.414 | BRANCH  | BRANCH  |

| id | p1.x, p1.y | p2.x, p2.y | length  | p1_type | p2_type |
|----|------------|------------|---------|---------|---------|
| 59 | [44,211]   | [53,224]   | 16.899  | BRANCH  | BRANCH  |
| 60 | [16,203]   | [32,212]   | 18.728  | TIP     | BRANCH  |
| 61 | [34,215]   | [32,212]   | 2.828   | TIP     | BRANCH  |
| 62 | [72,244]   | [53,224]   | 30.385  | TIP     | BRANCH  |
| 63 | [127,257]  | [126,266]  | 8.414   | TIP     | BRANCH  |
| 64 | [136,266]  | [126,266]  | 9.0     | TIP     | BRANCH  |
| 65 | [124,266]  | [126,266]  | 1.0     | TIP     | BRANCH  |
| 66 | [198,261]  | [213,267]  | 16.485  | TIP     | BRANCH  |
| 67 | [216,261]  | [213,267]  | 5.828   | TIP     | BRANCH  |
| 68 | [213,267]  | [245,286]  | 40.213  | BRANCH  | BRANCH  |
| 69 | [248,284]  | [245,286]  | 2.828   | TIP     | BRANCH  |
| 70 | [250,293]  | [245,286]  | 8.071   | TIP     | BRANCH  |
| 71 | [71,329]   | [59,305]   | 29.971  | TIP     | BRANCH  |
| 72 | [59,305]   | [53,309]   | 6.243   | BRANCH  | BRANCH  |
| 73 | [53,309]   | [53,314]   | 4.0     | BRANCH  | BRANCH  |
| 74 | [0,262]    | [53,309]   | 74.569  | TIP     | BRANCH  |
| 75 | [55,314]   | [53,314]   | 1.0     | TIP     | BRANCH  |
| 76 | [53,314]   | [48,319]   | 6.243   | BRANCH  | BRANCH  |
| 77 | [370,330]  | [381,318]  | 17.314  | TIP     | BRANCH  |
| 78 | [379,316]  | [381,318]  | 1.414   | TIP     | BRANCH  |
| 79 | [384,316]  | [381,318]  | 2.414   | TIP     | BRANCH  |
| 80 | [36,323]   | [48,319]   | 12.657  | TIP     | BRANCH  |
| 81 | [47,326]   | [48,319]   | 6.414   | TIP     | BRANCH  |
| 82 | [361,338]  | [354,341]  | 7.243   | TIP     | BRANCH  |
| 83 | [358,320]  | [354,341]  | 26.142  | TIP     | BRANCH  |
| 84 | [358,366]  | [354,341]  | 25.657  | TIP     | BRANCH  |
| 85 | [10,371]   | [18,378]   | 11.071  | TIP     | BRANCH  |
| 86 | [18,380]   | [18,378]   | 1.0     | TIP     | BRANCH  |
| 87 | [20,375]   | [18,378]   | 2.414   | TIP     | BRANCH  |
| 88 | [314,380]  | [313,382]  | 1.414   | TIP     | BRANCH  |
| 89 | [301,385]  | [313,382]  | 13.071  | TIP     | BRANCH  |
| 90 | [313,382]  | [353,464]  | 103.154 | BRANCH  | BRANCH  |

| id  | p1.x, p1.y | p2.x, p2.y | length | p1_type | p2_type |
|-----|------------|------------|--------|---------|---------|
| 91  | [414,401]  | [422,438]  | 39.314 | TIP     | BRANCH  |
| 92  | [432,440]  | [422,438]  | 9.828  | TIP     | BRANCH  |
| 93  | [422,438]  | [378,459]  | 51.284 | BRANCH  | BRANCH  |
| 94  | [69,476]   | [20,444]   | 63.012 | TIP     | BRANCH  |
| 95  | [18,447]   | [20,444]   | 2.414  | TIP     | BRANCH  |
| 96  | [378,459]  | [353,464]  | 26.071 | BRANCH  | BRANCH  |
| 97  | [381,470]  | [378,459]  | 10.828 | TIP     | BRANCH  |
| 98  | [353,464]  | [277,481]  | 84.941 | BRANCH  | BRANCH  |
| 99  | [148,466]  | [126,475]  | 24.728 | TIP     | BRANCH  |
| 100 | [126,475]  | [121,491]  | 17.071 | BRANCH  | BRANCH  |
| 101 | [125,472]  | [126,475]  | 2.0    | TIP     | BRANCH  |
| 102 | [254,450]  | [275,480]  | 37.698 | TIP     | BRANCH  |
| 103 | [275,480]  | [277,487]  | 29.314 | BRANCH  | BRANCH  |
| 104 | [275,480]  | [277,481]  | 1.0    | BRANCH  | BRANCH  |
| 105 | [277,481]  | [277,487]  | 5.0    | BRANCH  | BRANCH  |
| 106 | [285,508]  | [277,487]  | 23.314 | TIP     | BRANCH  |
| 107 | [116,491]  | [121,491]  | 4.0    | TIP     | BRANCH  |
| 108 | [133,510]  | [121,491]  | 23.556 | TIP     | BRANCH  |
| 109 | [378,0]    | [370,21]   | 25.142 | TIP     | TIP     |
| 110 | [405,3]    | [398,22]   | 31.657 | TIP     | TIP     |
| 111 | [337,7]    | [323,13]   | 16.485 | TIP     | TIP     |
| 112 | [480,17]   | [508,17]   | 30.485 | TIP     | TIP     |
| 113 | [453,23]   | [432,34]   | 25.556 | TIP     | TIP     |
| 114 | [437,47]   | [420,48]   | 17.414 | TIP     | TIP     |
| 115 | [510,49]   | [456,87]   | 88.184 | TIP     | TIP     |
| 116 | [192,75]   | [176,79]   | 17.657 | TIP     | TIP     |
| 117 | [309,76]   | [322,89]   | 21.314 | TIP     | TIP     |
| 118 | [345,82]   | [378,147]  | 79.255 | TIP     | TIP     |
| 119 | [232,94]   | [233,107]  | 15.071 | TIP     | TIP     |
| 120 | [381,101]  | [389,101]  | 8.828  | TIP     | TIP     |
| 121 | [323,102]  | [324,111]  | 10.243 | TIP     | TIP     |
| 122 | [443,139]  | [439,156]  | 18.657 | TIP     | TIP     |

| id  | p1.x, p1.y | p2.x, p2.y | length | p1_type | p2_type |
|-----|------------|------------|--------|---------|---------|
| 123 | [159,142]  | [160,160]  | 20.899 | TIP     | TIP     |
| 124 | [368,152]  | [355,186]  | 39.385 | TIP     | TIP     |
| 125 | [268,153]  | [264,175]  | 28.485 | TIP     | TIP     |
| 126 | [386,167]  | [395,174]  | 13.071 | TIP     | TIP     |
| 127 | [420,167]  | [412,174]  | 10.899 | TIP     | TIP     |
| 128 | [92,182]   | [111,204]  | 34.456 | TIP     | TIP     |
| 129 | [239,186]  | [207,201]  | 38.213 | TIP     | TIP     |
| 130 | [485,215]  | [509,230]  | 30.799 | TIP     | TIP     |
| 131 | [143,226]  | [131,245]  | 35.385 | TIP     | TIP     |
| 132 | [243,229]  | [224,245]  | 26.799 | TIP     | TIP     |
| 133 | [275,237]  | [270,253]  | 18.071 | TIP     | TIP     |
| 134 | [34,257]   | [37,272]   | 18.828 | TIP     | TIP     |
| 135 | [74,260]   | [87,274]   | 21.971 | TIP     | TIP     |
| 136 | [468,281]  | [454,286]  | 16.071 | TIP     | TIP     |
| 137 | [392,288]  | [376,291]  | 17.243 | TIP     | TIP     |
| 138 | [285,289]  | [299,301]  | 18.971 | TIP     | TIP     |
| 139 | [493,312]  | [464,321]  | 32.728 | TIP     | TIP     |
| 140 | [73,315]   | [79,321]   | 8.485  | TIP     | TIP     |
| 141 | [170,327]  | [181,336]  | 14.728 | TIP     | TIP     |
| 142 | [499,344]  | [504,364]  | 22.071 | TIP     | TIP     |
| 143 | [447,345]  | [441,373]  | 32.485 | TIP     | TIP     |
| 144 | [132,362]  | [139,377]  | 19.314 | TIP     | TIP     |
| 145 | [269,366]  | [270,378]  | 12.414 | TIP     | TIP     |
| 146 | [206,383]  | [190,388]  | 18.071 | TIP     | TIP     |
| 147 | [59,391]   | [79,392]   | 22.899 | TIP     | TIP     |
| 148 | [477,412]  | [490,433]  | 26.385 | TIP     | TIP     |
| 149 | [231,429]  | [232,437]  | 8.414  | TIP     | TIP     |
| 150 | [462,470]  | [455,481]  | 13.899 | TIP     | TIP     |
| 151 | [183,474]  | [201,482]  | 21.899 | TIP     | TIP     |
| 152 | [382,476]  | [383,488]  | 12.414 | TIP     | TIP     |
| 153 | [240,487]  | [235,494]  | 9.657  | TIP     | TIP     |
| 154 | [153,493]  | [157,509]  | 18.485 | TIP     | TIP     |

| id  | p1.x, p1.y | p2.x, p2.y | length | p1_type | p2_type |
|-----|------------|------------|--------|---------|---------|
| 155 | [12,500]   | [21,500]   | 10.657 | TIP     | TIP     |